DeCODER distribution file-type assay

Estimated from URL text on schema:DataDownload | 8 October 2026

Each catalog was queried on its own endpoint. Counts are not added across catalogs. A distribution is a schema:DataDownload. The file-type estimate uses schema:contentUrl only, not schema:url and not schema:encodingFormat.

Query

```
PREFIX schema: <https://schema.org/>
SELECT DISTINCT ?url WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          schema:contentUrl ?url .
  }
}
LIMIT 200000
```

How a URL was typed

The path is taken before the query string and percent-decoded. If the last path segment ends in a known file suffix, that suffix is the type. Otherwise a query-parameter value is used when that value is itself a filename (this is how Nextcloud links such as files=tile.tif are typed). Compound names such as .csv.gz stay compound. Service scripts (.php, .do, .exe) are not treated as file types. Anything else is unknown. This is an estimate from the URL string. The bytes were not downloaded.

What was in the graphs

Catalog	Distinct contentUrl	DataDownload objects
deepoceans	33,744	154,333
ecoforecast	1,122	2,022
earthsurface	5,211	5,331
facetsearch	50,274	147,554

Endpoints: deepoceans, ecoforecast, and earthsurface on https://qlever.geocodes-aws.earthcube.org/graphspace/<name> ; facetsearch on https://qlever.geocodes-aws-dev.earthcube.org/graphspace/facetsearch. Object counts are higher than URL counts because many downloads share a URL.

deepoceans

https://qlever.geocodes-aws.earthcube.org/graphspace/deepoceans

Type from URL text	Distinct URLs	Share
unknown	14,254	42.2%
.csv	8,456	25.1%
.zip	5,905	17.5%
.nc	3,648	10.8%
.txt	1,054	3.1%
other recognized suffix	427	1.3%

Other suffixes are mostly .html, .gz, .pdf, .xlsx, .rar, .xml, and .tgz. No URL contains parquet. Unknown is mostly dataset pages with no filename (including 7,431 on www.bco-dmo.org), IPT archive.do links (5,531), and WMS/OWS endpoints (462).

ecoforecast

https://qlever.geocodes-aws.earthcube.org/graphspace/ecoforecast

Type from URL text	Distinct URLs	Share
unknown	695	61.9%
.json	403	35.9%
.r	17	1.5%
.rmd	5	0.4%
.csv	2	0.2%

No contentUrl ends in .parquet. Of the unknown URLs, 354 are GitHub or STAC browser pages, 220 are S3 paths with no file suffix, and 117 are S3 paths whose text contains bundled-parquet but still has no .parquet suffix. encodingFormat is application/x-parquet for 337 of these URLs, so the URL text underestimates parquet.

earthsurface

https://qlever.geocodes-aws.earthcube.org/graphspace/earthsurface

Type from URL text	Distinct URLs	Share
.tif	4,609	88.4%
.gpkg	460	8.8%
unknown	123	2.4%
other recognized suffix	19	0.4%

Almost all .tif (4,601) and the .gpkg names sit in the query string (files=...), not the path. The other suffixes are .txt (6), .html (5), .zip (5), .nc, .pdf, and .xml. Unknown is share links, directories, and DOI landing pages. No parquet.

facetsearch

https://qlever.geocodes-aws-dev.earthcube.org/graphspace/facetsearch

Type from URL text	Distinct URLs	Share
unknown	34,113	67.9%
.zip	5,704	11.3%
.txt	5,497	10.9%
.nc	3,571	7.1%
.csv	1,236	2.5%
other recognized suffix	153	0.3%

Unknown is mostly IODP secureGetObj.php?id=... (26,155) and IPT archive.do links (5,437), plus WMS/OWS (285) and DOI pages (159). No parquet. This graph overlaps some ocean URLs; it was not joined with deepoceans.

Reading the unknown bucket

unknown means the URL text has no recognized filename suffix. It is not a claim that the bytes have no format. IPT archive.do links are download endpoints whose URLs do not say .zip. IODP secureGetObj.php links are opaque ids. S3 forecast paths often omit .parquet even when encodingFormat says parquet. A directory URL or a DOI is a landing place, not a file suffix.